Task 1: S3 read-only policy for playlistViewer

Create the following customer-managed policy in IAM → Policies → Create policy → JSON.

Policy name: PlaylistBucketReadOnly

This policy permits listing the specified bucket and reading its objects, but not uploading or deleting them.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ListMusicPlaylistsBucket",
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::my-music-playlists"
    },
    {
      "Sid": "ReadMusicPlaylistObjects",
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-music-playlists/*"
    }
  ]
}
```

Procedure:
1. Create the policy using the JSON above.
2. Create a new IAM user named playlistViewer.
3. Attach the PlaylistBucketReadOnly policy to the user.
4. Save the policy and user details.

Task 2: Inline policy for devOpsIntern

Create an inline policy directly on the existing user devOpsIntern.

Policy name: DevTaggedEC2StartStop

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "StartStopDevInstancesOnly",
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "arn:aws:ec2:*:*:instance/*",
      "Condition": {
        "StringEquals": {
          "ec2:ResourceTag/Environment": "Dev"
        }
      }
    },
    {
      "Sid": "DescribeInstances",
      "Effect": "Allow",
      "Action": "ec2:DescribeInstances",
      "Resource": "*"
    }
  ]
}
```

Explanation:

- ec2:StartInstances allows starting instances.

- ec2:StopInstances allows stopping instances.
- The condition restricts these actions to instances tagged Environment=Dev.
- ec2:DescribeInstances supports viewing instance information; this action requires "Resource": "*".

Procedure: Open IAM → Users → devOpsIntern → Permissions → Add permissions → Create inline policy. Paste the JSON, review it, and save.

Task 3: DynamoDB read-only managed policy

Create a customer-managed policy named DynamoDBTableViewer.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ViewDynamoDBTables",
      "Effect": "Allow",
      "Action": [
        "dynamodb:ListTables",
        "dynamodb:DescribeTable"
      ],
      "Resource": "*"
    }
  ]
}
```

Procedure:
1. Open IAM → Policies → Create policy.
2. Paste the JSON and create DynamoDBTableViewer.
3. Open User groups → appReviewers.
4. Attach the policy to the group.

Task 4: Fix the S3 policy error

The supplied policy allows reading objects but does not allow listing the bucket. Add a separate statement for s3:ListBucket.
Corrected policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ListUserUploads",
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::user-uploads"
    },
    {
      "Sid": "ReadUserUploadObjects",
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::user-uploads/*"
    }
  ]
}
```

Task 5: DynamoDB policy to update only the user's own profile

Policy name: UpdateOwnUserProfile

The following policy permits a user to update their own profile record in a DynamoDB table named UserProfiles.

Assumption: The table's partition key is username, and its value matches the IAM user's username. The condition must match the actual table key design for this policy to work as intended.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "UpdateOwnProfileOnly",
      "Effect": "Allow",
      "Action": "dynamodb:UpdateItem",
      "Resource":
"arn:aws:dynamodb:ap-south-1:123456789012:table/UserProfiles",
      "Condition": {
        "ForAllValues:StringEquals": {
          "dynamodb:LeadingKeys": [
            "${aws:username}"
          ]
        }
      }
    }
  ]
}
```

Before using this policy:
- Replace 123456789012 with your AWS account ID.
- Change the region if your table is not in Mumbai (ap-south-1).
- Ensure the table's partition key and IAM username mapping match the policy.
- Test it in the AWS IAM Policy Simulator and, ideally, in a test environment.